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Department of Artificial Intelligence

Machine Learning (IA206)

LAB ASSIGNMENT

Objectives:

- Understand the concept of simple and multiple linear regression.
- Perform exploratory data analysis (EDA) before modeling.
- Train and evaluate linear regression models.
- Interpret coefficients and residuals.

Part A: Exploratory Data Analysis

1. Load the dataset (e.g., Boston Housing or any regression dataset).
2. Plot the distribution of the target variable. Comment on skewness.
3. Compute and visualize the correlation matrix (heatmap). Which features aremost correlated with the target?

Part B: Simple Linear Regression

1. Select one predictor
2. Formulate the hypothesis equation:$y=\beta 0+\beta 1x+\epsilon$
3. Fit a simple linear regression model.
4. Report:
    - Intercept ($\beta 0$)
    - Slope ($\beta 1$)
5. Plot scatter plot of feature vs target with the regression line.
6. Interpret the slope: What does a unit increase in x mean for y?

Part C: Multiple Linear Regression

1. Select at least 3 predictor variables.
2. Fit a multiple linear regression model. Write down the fitted equation.
3. Report model coefficients and interpret any one of them.
4. Evaluate the model using:
    - R^2 score
    - Mean Squared Error (MSE)
    - Root Mean Squared Error (RMSE)
5. Compare performance with the simple regression model. Which one is better? Why?